

FITBUDDY

AI Fitness Plan Generator using Gemini Models

PROJECT DOCUMENTATION PHASE

1. Introduction

FitBuddy is an AI-powered fitness plan generation application designed to help users create personalized workout routines. The system uses Gemini models to process user requirements and generate a structured fitness plan in a simple and readable format.

2. Project Background

People often find it difficult to create a workout routine that matches their personal goals, experience level, available time, and preferences. Generic workout plans may not be suitable for every user. FitBuddy is proposed as a technology-based solution that uses generative AI to assist users in creating a customized starting plan.

3. Problem Definition

The main problem is the lack of a simple and personalized way for users to organize their fitness routines. Users may be unsure about which exercises to choose, how to arrange them across a week, and how to adjust the routine according to their available time and experience.

4. Objectives

- Develop an AI-based fitness plan generator.
- Collect useful user information such as fitness goal, experience level, available time, and preferences.
- Use Gemini models to generate a structured workout plan.
- Present the generated plan in a clear and easy-to-read format.
- Create a simple and user-friendly application flow.
- Provide a foundation for future features such as progress tracking and plan regeneration.

5. Scope of the Project

The initial scope focuses on collecting user requirements, generating a personalized fitness plan through Gemini, and displaying the generated result. Advanced features such as wearable-device integration, medical assessment, and professional coaching are outside the initial scope and can be considered for future development.

PAGE 2 – REQUIREMENTS DOCUMENTATION

6. Functional Requirements

- The application should allow users to enter their fitness goal.
- The application should collect the user's experience level.
- The application should accept available workout days and time.
- The application should collect relevant workout preferences.
- The system should prepare the collected information for AI processing.
- The Gemini model should generate a structured fitness plan.
- The application should display the generated plan clearly.
- The system should handle missing or invalid required inputs.

7. Non-Functional Requirements

- **Usability:** The interface should be simple enough for a first-time user.
- **Performance:** The system should process requests within a reasonable response time.
- **Reliability:** The application should handle common input and API errors properly.
- **Maintainability:** Code and project components should be organized for future changes.
- **Security:** API credentials and user information should be handled securely.
- **Scalability:** The design should allow additional fitness features to be added later.

8. User Input Requirements

Input	Description
Fitness Goal	Target selected by the user, such as general fitness or strength improvement.
Experience Level	Beginner, intermediate, or advanced level.
Workout Frequency	Number of available workout days.
Available Time	Approximate time available for each session.
Workout Preference	Preferred type or style of workout.
Equipment	Available equipment or home/gym preference.

9. Expected Output

The output should contain an organized workout schedule with suitable exercise suggestions, approximate session structure, and simple instructions. The generated result should be formatted so that users can easily understand the plan.

PAGE 3 – SYSTEM DESIGN DOCUMENTATION

10. System Architecture – High Level

User Interface → Input Collection → Validation → Prompt Preparation → Gemini API/Model → AI Generated Plan → Response Processing → Fitness Plan Display

11. Main Components

- **User Interface:** Collects user information and displays the generated plan.
- **Input Validation:** Checks whether required information has been provided correctly.
- **Prompt Builder:** Converts user inputs into a structured request for the Gemini model.
- **Gemini Model:** Generates the fitness plan based on the supplied requirements.
- **Response Handler:** Receives and processes the model response.
- **Plan Display Module:** Formats the generated result for the user.

12. Processing Flow

Step	Process
1	User Input
2	Validate Required Information
3	Prepare AI Prompt
4	Send Request to Gemini
5	Receive Generated Response
6	Format Fitness Plan
7	Display Result

13. Sample Prompt Structure

Role: Fitness planning assistant.
Inputs: Goal, experience level, available days, workout duration, preferences, and equipment.
Task: Generate a structured weekly fitness plan using the supplied information.
Output: Clear day-wise workout schedule with simple exercise guidance.

14. Data Flow

The user's information is collected by the application and converted into an AI request. The Gemini model processes the request and returns generated text. The application then formats the response and presents the fitness plan to the user.

PAGE 4 – IMPLEMENTATION & TESTING DOCUMENTATION

15. Technology Requirements

Technology	Purpose
Gemini Models / API	AI-based fitness plan generation.
Frontend	Collect inputs and display results.
Backend/Application Logic	Validation, prompt creation, API communication, and response handling.
Programming Language	Can be selected based on the project implementation, such as Python or JavaScript.
Internet/API Access	Required for communication with the Gemini service when using a cloud API.

16. Testing Plan

Test Case	Expected Result
Valid user inputs	A structured fitness plan is generated.
Missing goal	System asks the user to provide the required goal.
Missing experience level	System requests the missing information.
Invalid time value	System rejects or requests correction of the value.
API/model error	System shows an appropriate error message instead of crashing.
Long generated response	Response is formatted for readable display.

17. Error Handling

- Validate important fields before sending the request.
- Handle API failures with a user-friendly message.
- Do not expose secret API keys in the frontend or public source code.
- Provide a retry or regenerate option where appropriate.
- Prevent empty or incomplete requests from being sent.

18. Security Considerations

API keys should be stored securely and not hard-coded into publicly shared client-side code. Only necessary user information should be collected. The project should avoid requesting sensitive medical information unless a future version has an appropriate privacy and safety design.

PAGE 5 – DOCUMENTATION SUMMARY & FUTURE SCOPE

19. Project Deliverables

- Working FitBuddy application/interface.
- User input form for fitness preferences.
- Gemini-based AI generation component.
- Generated fitness-plan display.
- Project source code and configuration documentation.
- Testing documentation and sample outputs.
- Project presentation/demo materials.

20. Limitations

- AI-generated plans may not be suitable for every individual.
- Generated content can require user review.
- The quality of the result depends partly on the quality of user inputs and prompt design.
- Internet access may be required when using a cloud-based Gemini API.
- The initial version is not intended to diagnose medical conditions or replace professional advice.

21. Future Enhancements

- Workout progress tracking.
- Personalized plan regeneration based on progress.
- Exercise video or demonstration library.
- Workout reminders and notifications.
- Weekly fitness dashboard.
- Multilingual support.
- Optional integration with fitness/activity data.
- More detailed personalization based on user feedback.

22. Conclusion

This documentation defines the purpose, requirements, architecture, processing flow, implementation considerations, testing approach, limitations, and future scope of FitBuddy. The project demonstrates how Gemini-based generative AI can be applied to create a personalized fitness planning experience. The documentation can serve as a reference during development, testing, presentation, and future enhancement.

Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models

Phase: Project Documentation Phase

Document: 5-Page Project Documentation